SWASTHYAGRID

AI-Powered Health Resource Resilience Network // User Walkthrough Guide

Executive Purpose: SwasthyaGrid is a unified public health intelligence and logistics resilience platform designed for India's Primary Health Centres (PHCs), Community Health Centres (CHCs), and District Hospitals. It delivers real-time visibility into medicine stocks, bed occupancy, and healthcare workforce availability, automating inter-facility resource balancing to prevent stockouts and manage emergency health surges.

1. Introduction & Operational Rationale

India's public healthcare delivery is anchored by over 30,000 rural and urban health centres. SwasthyaGrid solves critical structural operational bottlenecks:

- **Information Silos:** Health authorities historically detect stockouts only after facilities exhaust essential drugs. SwasthyaGrid establishes real-time telemetry across all tiers.
- **Surplus-Deficit Mismatch:** Resolves localized shortages by detecting neighbouring centres with surplus inventory and routing buffer stock dynamically.
- **Proactive Surge Preparedness:** Applies machine learning to forecast monsoon disease surges (e.g. Dengue, Malaria, Diarrhoea) 14 days in advance.
- **Zero Expiry Waste:** Automatically flags batches nearing expiration within 90 days, facilitating prompt redistribution to high-consumption hubs.

OPERATIONAL CHALLENGE	TRADITIONAL WORKFLOW	SWASTHYAGRID INTEGRATED PLATFORM
Stockout Identification	Discovered during patient consultation	AI early warning 3-7 days in advance
Facility Coordination	Isolated, independent local management	Unified grid sharing surplus stocks
Bed Census & Surge	Manual phone calls and paper logs	Real-time general, ICU and O2 bed tracking
Logistics Dispatch	Slow administrative approvals (days)	Automated digital transfer requisition (< 1h)

2. System Overview & Core Modules

MODULE	KEY CAPABILITIES	ADMINISTRATIVE VALUE
Executive Dashboard	Macro KPI cards, bed census charts, critical risk alerts, live transfer status.	Instant executive situational awareness.
PHC Network	Digital directory of all PHCs, CHCs, SDHs and DHs with contact, GPS & beds.	100% facility asset and contact mapping.
Inventory & Medicines	Essential Drug List (EDL) compliance, supply runways (days), batch expiry tracking.	Eliminates emergency medicine stockouts.
Bed & ICU Surge	Live occupancy rates for general, ICU and oxygen-supported beds.	Surge capacity routing & divert planning.
Staff & Attendance	Clinical rosters (Doctors, Nurses, ANMs, Pharmacists) and shift attendance.	Enforces 100% clinical doctor coverage.
AI Forecasting	14-day outpatient footfall and critical drug consumption projections.	Enables proactive buffer stocking.
Risk Early Warnings	Automated anomaly detection (Critical, High, Moderate) with AI action prompts.	Prioritizes urgent interventions.
Resource Transfers	Inter-facility stock balancing with digital dispatch, approval, and ETA tracking.	Rapid mutual-aid supply logistics.
Audit Logs	Immutable compliance ledger of all user actions, logins, and authorizations.	Complete governance and accountability.

3. Administrative User Roles & Hierarchy

USER ROLE	ADMINISTRATIVE SCOPE	PRIMARY RESPONSIBILITIES
National Administrator	Nationwide Health Network	Macro KPIs, inter-state health corridor monitoring, national central buffer allocation.
State Administrator	State-Wide (All Districts)	State buffer depots, inter-district resource transfers, regional outbreak surge management.
District Administrator (CDMO)	District Network (PHCs & CHCs)	Authorize local resource transfers, review daily census, resolve staffing and stock deficits.
PHC Administrator (MO / Store)	Local Health Facility	Daily OPD census entry, medicine inward verification, batch monitoring, transfer requests.

4. Step-by-Step Operational Routine (7 Steps)

Step 1: Secure Login: Access http://localhost:3000 with authorized government credentials.

Step 2: Executive Dashboard Inspection: Open Dashboard to review 7 top health indicators and live grid sync.

Step 3: Review Health Indicators: Audit facilities mapped, active PHCs, drug availability (%), and bed occupancy.

Step 4: Triage Critical Risk Alerts: Review Red/Amber early warnings (e.g. stockout predicted in < 36 hours).

Step 5: Audit Medicine Inventory: Inspect items with < 7 days supply runway and check cold-chain 2°C-8°C biologicals.

Step 6: Gap Analysis: Identify nearest health centres holding surplus stock of the deficit item.

Step 7: Execute Resource Reallocation: Initiate digital transfer from surplus source to target PHC in one click.

5. Dashboard Metrics & Indicator Thresholds

INDICATOR	BENCHMARK	OPERATIONAL MEANING	ACTION THRESHOLD
Total Facilities	1,248 Units	Total mapped public health facilities in grid	Maintain 100% digital telemetry
Active PHCs	892 Units (98.6%)	Health centres actively reporting daily OPD	Flag if active rate drops < 95%
Medicine Availability	92.4%	Essential Drug List (EDL) in-stock ratio	Target >= 90%. Action below 85%
Bed Availability	74.8% (4,820 Ready)	Occupancy load across general & ICU wards	Prepare surge divert if > 85%
Staff Strength	6,420 (91.4%)	Registered doctors, nurses, ANMs on duty	Audit rosters with attendance < 90%
Critical Alerts	3 Active	Imminent stockouts or outbreak surges	Action within 24 hours mandatory
Pending Transfers	8 En-Route	Active inter-facility logistics shipments	Monitor deliveries taking > 6h

6. Practical User Walkthrough Scenarios

Scenario A: Resolving a Critical Medicine Stockout Threat

1. **Alert Trigger:** District dashboard flags Red alert: *'Tangi CHC Amoxicillin stockout in < 36 hrs (80 vials remain)'*.
2. **Automated Discovery:** System scans regional inventory and identifies *SCB Medical College Store* with 4,200 surplus vials (35 km away).
3. **Requisition Approval:** CDMO authorizes transfer of 1,000 vials. Depot confirms packing; status updates to **In Transit**.
4. **Delivery & Verification:** Tangi CHC scans batch QR code on arrival. Inventory increments and alert auto-resolves.

Scenario B: Emergency Bed Surge Management During Outbreak

1. **Outbreak Flag:** Dengue cluster detected in Gop Block; *Konark PHC* reaches 93.3% bed occupancy (28/30 beds).
2. **Cluster Capacity Scan:** District Officer reviews Bed Management map, finding *Pipili PHC* (22 km away) at 58% occupancy with 10 vacant beds.
3. **Proactive Divert:** 108 emergency ambulance dispatch protocol temporarily diverts non-critical admissions to Pipili PHC.

Scenario C: Weekend Specialist & Nursing Deficit Resolution

1. **Gap Detection:** Staff module flags zero anesthetist coverage for upcoming Saturday night shift at *Soro SDH*.
2. **Reserve Pool Lookup:** System finds 2 available specialist officers on reserve duty in the Balasore District Headquarters pool.
3. **Digital Assignment:** Administrator issues temporary roster reassignment, transmitting automated SMS confirmation.

7. Administrative Best Practice Guidelines

TIMING	PROTOCOL	KEY OBJECTIVE
08:30 IST (Morning)	Review top KPI cards; triage Red critical alerts.	Prevent same-day stockouts and clinical disruptions.
14:00 IST (Mid-Day)	Authorize inter-facility transfer slips; audit inward arrivals.	Ensure timely logistics dispatch before road peak hours.
17:30 IST (Evening)	Verify bed occupancy updates; audit clinical duty rosters.	Guarantee 100% emergency night-shift coverage.
Weekly Audit	Review 14-day AI forecast models and 90-day expiry schedules.	Zero medicine expiration waste across all district depots.

8. Frequently Asked Questions (FAQ)

Q1: What is SwasthyaGrid?
A: An AI-powered public health resilience platform unifying Primary Health Centres, Community Health Centres, and District Hospitals into a coordinated resource tracking and mutual-aid sharing network.

Q2: Who is authorized to access SwasthyaGrid?
A: Public health personnel including National Health Mission directors, State Health Secretaries, Chief District Medical Officers (CDMOs), PHC Medical Officers, Hospital Superintendents, and medical storekeepers.

Q3: What data does the platform manage?
A: Essential medicine inventory, batch expiries, cold-chain telemetry (2-8°C), general & ICU bed census, clinical duty rosters, epidemiological outbreak indicators, and inter-facility logistics.

Q4: How does SwasthyaGrid support decision-making?
A: Replaces slow manual reports with real-time telemetry, automated risk alerts, and prescriptive AI recommendations that calculate exact transfer amounts to avert shortages.

Q5: Can the platform scale across multiple states?
A: Yes. SwasthyaGrid is architected on a high-concurrency microservice foundation designed to connect over 30,000 health centres nationwide across all states and union territories.

National Health Systems Helpdesk: Email: support@swasthya-grid.gov.in | Toll-Free Emergency Helpline: 1800-11-HEALTH | National Health Mission Integrated Operations Command